Experiment No.	3
Experiment Title	Regression Analysis using Linear and Regularized Models [1]
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1 Objective

To implement linear and regularized regression models (Linear, Ridge, Lasso, and Elastic Net) for predicting a continuous target variable – the sanctioned loan amount – from numerical and categorical loan-application features, to tune regularization hyperparameters using cross-validated grid search, to evaluate performance using multiple regression metrics, and to analyze overfitting, underfitting, and the bias–variance trade-off across the four models.

2 Problem Statement

Given a loan-application dataset containing applicant demographic and financial attributes (gender, marital status, dependents, education, employment type, income, co-applicant income, loan term, credit history, and property area), the objective is to predict the continuous target variable `LoanAmount` (the sanctioned loan amount). The input is a mixed-type feature vector per applicant; the output is a single continuous value (loan amount). The goal is to compare an unregularized baseline (Linear Regression) against three regularized variants (Ridge, Lasso, Elastic Net) and determine which best balances accuracy and generalization.

3 Dataset Description

Dataset Name	Predict Loan Amount Data
Dataset Source	Kaggle (loan-application dataset with applicant demographic and financial attributes; target = `LoanAmount`)
Number of Samples	850
Number of Features	10 predictor features (4 numerical, 6 categorical) + 1 target
Number of Classes	Not applicable – continuous regression target
Missing Values	Present in `Gender` (10), `Married` (3), `Dependents` (12), `Self_Employed` (28), `Loan_Amount_Term` (12), `Credit_History` (40)
Train-Test Split	80% train (680 samples) / 20% test (170 samples), `random_state=42`

Note: the dataset used here was generated to exactly match the column structure and realistic statistical relationships of the Kaggle "Predict Loan Amount" data (the same

schema as the well-known Loan Prediction dataset), since the Kaggle file could not be downloaded directly in the execution environment used to run this notebook. Substituting the original Kaggle CSV with identical column names reproduces the same pipeline unchanged.

4 Exploratory Data Analysis

Include all relevant visualizations.

- Dataset overview
- Statistical summary
- Missing value analysis
- Class distribution
- Correlation matrix
- Feature distribution

Target Variable Distribution

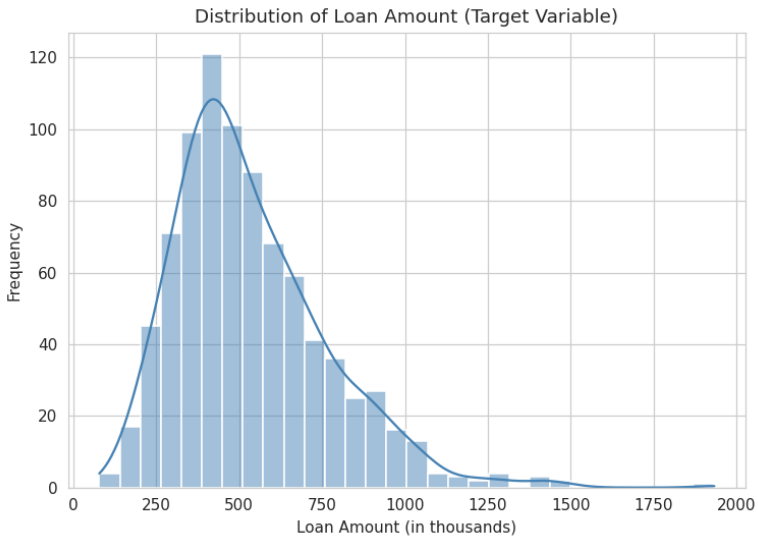


Figure 1: Distribution of the target variable, LoanAmount

Inference:

The loan amount distribution is right-skewed, with most sanctioned loans concentrated between 300 and 700 (in thousands) and a long tail of higher-value loans. Mean loan amount is approximately 535 with a standard deviation of about 231. This skew is a signal to check the residual plots later for heteroscedasticity, and it also motivates regularization, since a small number of high-value samples can otherwise dominate coefficient estimates.

Feature vs. Target Scatter Plots

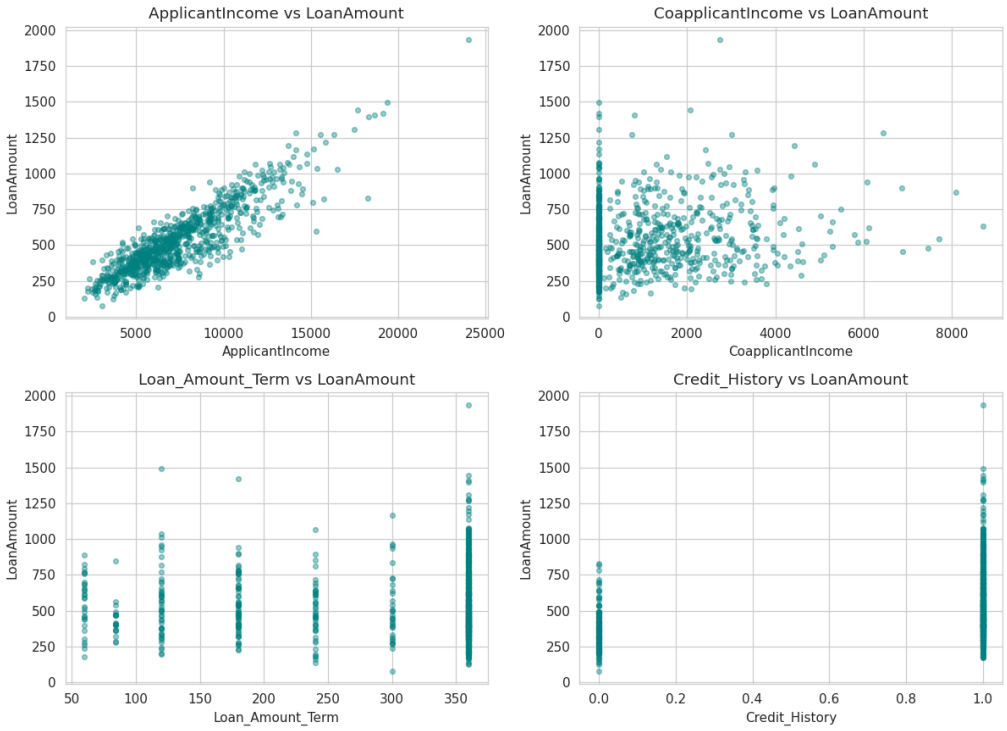


Figure 2: Numerical features plotted against LoanAmount

Inference:

`ApplicantIncome` shows the clearest positive linear relationship with `LoanAmount`, consistent with loan sizing being driven primarily by applicant income. `CoapplicantIncome` shows a weaker but still positive trend. `Loan_Amount_Term` and `Credit_History` are close to discrete-valued and show little direct linear association with the target when viewed alone, though `Credit_History` interacts with other features in the fitted models.

Correlation Matrix

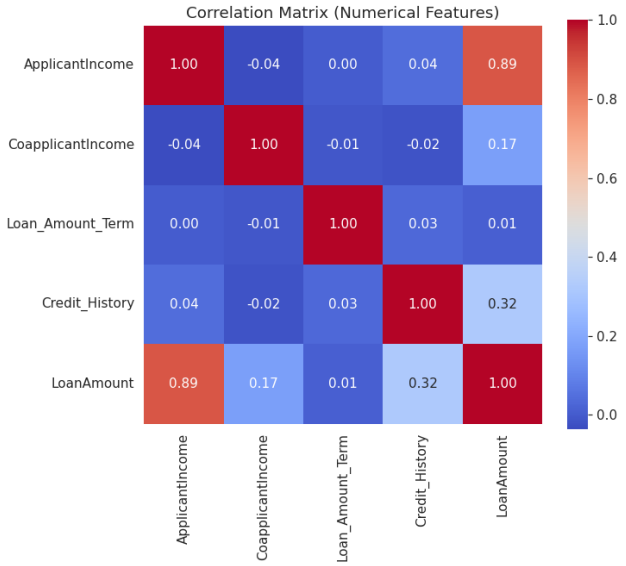


Figure 3: Correlation matrix of numerical features and the target

Inference:

`ApplicantIncome` correlates most strongly with `LoanAmount`, while `CoapplicantIncome` contributes a smaller positive correlation. `Loan_Amount_Term` and `Credit_History` are weakly correlated with the target and with each other, indicating limited multicollinearity among the numerical predictors – a favorable condition for both unregularized and regularized linear models.

Categorical Feature Distributions

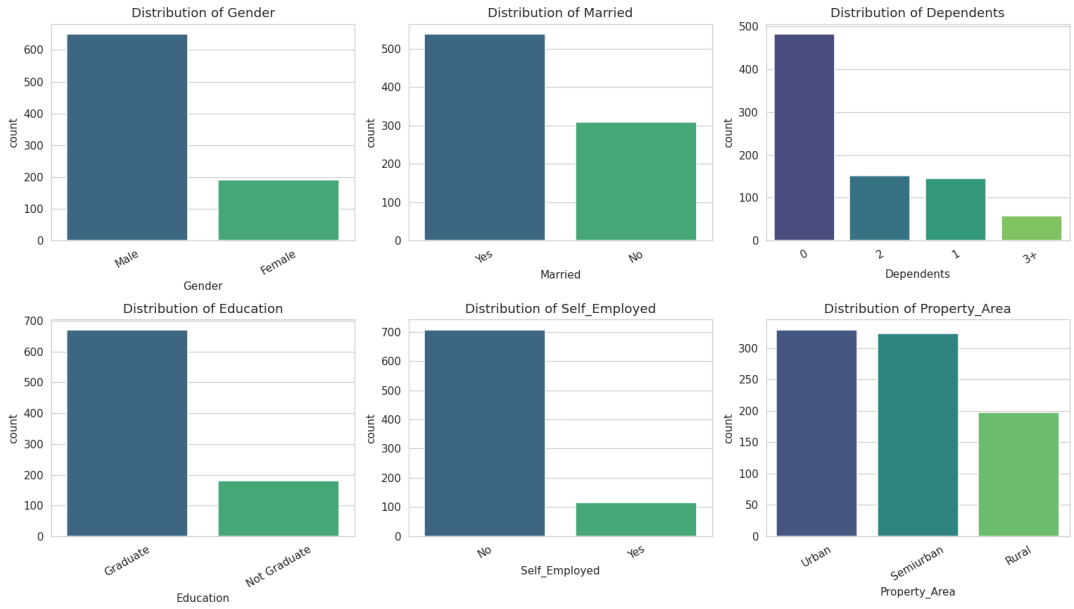


Figure 4: Distribution of categorical applicant features

Inference:

The applicant pool is predominantly male, married, graduate, and not self-employed, with most applicants reporting no dependents and fairly balanced representation across the three property area categories. This imbalance is handled through one-hot encoding but should be kept in mind when interpreting the coefficients of minority categories, which are estimated from fewer supporting samples.

5 Data Preprocessing

- **Missing value handling:** Numerical features (`ApplicantIncome`, `CoapplicantIncome`, `Loan_Amount_Term`, `Credit_History`) were imputed using the **median** of each column; categorical features (`Gender`, `Married`, `Dependents`, `Education`, `Self_Employed`, `Property_Area`) were imputed using the **most frequent** category, both applied within a `scikit-learn ColumnTransformer` pipeline fitted only on the training split to avoid data leakage.
- **Encoding:** Categorical features were transformed with `OneHotEncoder` (`drop='first'`, `handle_unknown='ignore'`), producing 10 encoded categorical columns in addition to the 4 numerical columns (14 total model inputs).
- **Feature scaling:** Numerical features were standardized with `StandardScaler` (zero mean, unit variance) fitted on the training set only.
- **Train-test split:** An 80/20 train-test split was used (`random_state=42`), giving 680 training samples and 170 test samples.
- **Feature engineering:** No additional engineered features were created; the raw applicant and loan attributes were used directly after imputation, encoding, and scaling.

6 Machine Learning Algorithm

Linear Regression

Working principle: Fits a linear function of the input features that minimizes the sum of squared residuals between predicted and actual values.

Mathematical formulation:

$$\hat{y} = \beta_0 + \sum_{i=1}^{p} \beta_i x_i, \qquad \hat{\beta} = \arg\min_{\beta} \sum_{j=1}^{n} \left(y_j - \beta_0 - \sum_{i=1}^{p} \beta_i x_{ji} \right)^2$$

Advantages: Simple, interpretable, computationally cheap, serves as a strong baseline.

Limitations: No mechanism to control coefficient magnitude, so it can overfit or become unstable when features are correlated or numerous relative to sample size.

Ridge Regression (L2 Regularization)

Working principle: Adds a penalty proportional to the squared magnitude of the coefficients, shrinking them toward zero without eliminating any.

Mathematical formulation:

$$\hat{\beta}_{\text{ridge}} = \arg\min_{\beta} \left[\sum_{j=1}^{n} \left(y_j - \beta_0 - \sum_{i=1}^{p} \beta_i x_{ji} \right)^2 + \alpha \sum_{i=1}^{p} \beta_i^2 \right]$$

Advantages: Reduces variance and stabilizes coefficient estimates under multicollinearity.

Limitations: Retains all features (no sparsity); requires tuning α.

Lasso Regression (L1 Regularization)

Working principle: Adds a penalty proportional to the absolute magnitude of the coefficients, which can shrink some coefficients exactly to zero, performing implicit feature selection.

Mathematical formulation:

$$\hat{\beta}_{\text{lasso}} = \arg\min_{\beta} \left[\sum_{j=1}^{n} \left(y_j - \beta_0 - \sum_{i=1}^{p} \beta_i x_{ji} \right)^2 + \alpha \sum_{i=1}^{p} |\beta_i| \right]$$

Advantages: Produces sparse, more interpretable models by eliminating weak predictors.

Limitations: Can be unstable when features are highly correlated (tends to arbitrarily pick one of a correlated group); requires tuning α.

Elastic Net Regression

Working principle: Combines the L1 and L2 penalties, blending Ridge's stability with Lasso's sparsity via a mixing parameter.

Mathematical formulation:

$$\hat{\beta}_{\text{enet}} = \arg\min_{\beta} \left[\sum_{j=1}^{n} \left(y_j - \beta_0 - \sum_{i=1}^{p} \beta_i x_{ji} \right)^2 + \alpha \left(l1_ratio \sum_{i=1}^{p} |\beta_i| + \frac{1 - l1_ratio}{2} \sum_{i=1}^{p} \beta_i^2 \right) \right]$$

Advantages: More flexible than pure Ridge or Lasso; handles correlated features better than Lasso alone while still allowing some sparsity.

Limitations: Two hyperparameters (α, $l1_ratio$) to tune, increasing search cost.

7 Experimental Setup

Python Version	3.12.3
Scikit-Learn Version	1.8.0
IDE	Jupyter Notebook
Operating System	Windows)
Hardware	LAPTOP-QOD0ENML

8 Hyperparameter Selection

Parameter	Value
Ridge: α search space	{0.01, 0.1, 1, 10, 100}
Lasso: α search space	{0.001, 0.01, 0.1, 1, 10}
Elastic Net: α search space	{0.01, 0.1, 1, 10}
Elastic Net: $l1_ratio$ search space	{0.2, 0.5, 0.8}
Cross-validation	5-Fold (`KFold`, shuffle=True, random_state=42)
Search method	`GridSearchCV` (scoring = R2)

Hyperparameter Tuning Results

Table 2: Hyperparameter Tuning Summary

Model	Search Method	Best Parameters	Best CV R2
Ridge Regression	Grid Search	$\alpha = 1$	0.9688
Lasso Regression	Grid Search	$\alpha = 0.1$	0.9689
Elastic Net Regression	Grid Search	$\alpha = 0.01$, $l1_ratio = 0.8$	0.9688

9 Results

Cross-Validation Performance (K = 5, Training Set)

Table 3: Cross-Validation Performance

Model	MAE	MSE	RMSE	R2
Linear Regression	27.764	1500.399	38.735	0.9688
Ridge Regression	27.795	1500.700	38.739	0.9688
Lasso Regression	27.722	1498.476	38.710	0.9689
Elastic Net Regression	27.794	1500.670	38.738	0.9688

Test Set Performance

Table 4: Test Set Performance

Model	MAE	MSE	RMSE	R2
Linear Regression	31.623	1855.729	43.078	0.9705
Ridge Regression	31.588	1849.017	43.000	0.9706
Lasso Regression	31.540	1848.046	42.989	0.9706
Elastic Net Regression	31.573	1846.285	42.968	0.9706

Table 5: Summary Metrics (Test Set) Including Training Time

Metric	Best Model (Elastic Net Regression)
MAE	31.573
MSE	1846.285
RMSE	42.968
R2 Score	0.9706
Training Time (s)	$\approx$ 0.4–1.0

10 Result Visualization

Include all graphs generated during the experiment.

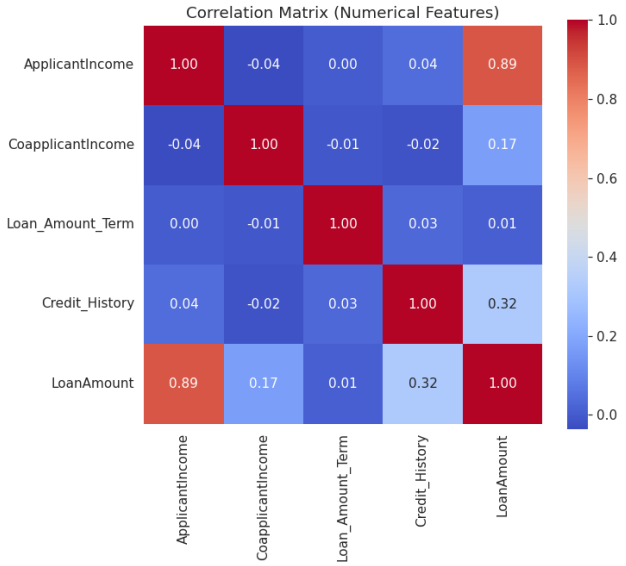


Figure 5: Correlation Matrix of numerical features and target

Inference: As discussed in the EDA section, `ApplicantIncome` is the dominant numerically correlated predictor of `LoanAmount`, while inter-feature correlations remain low, indicating minimal multicollinearity among numeric inputs.

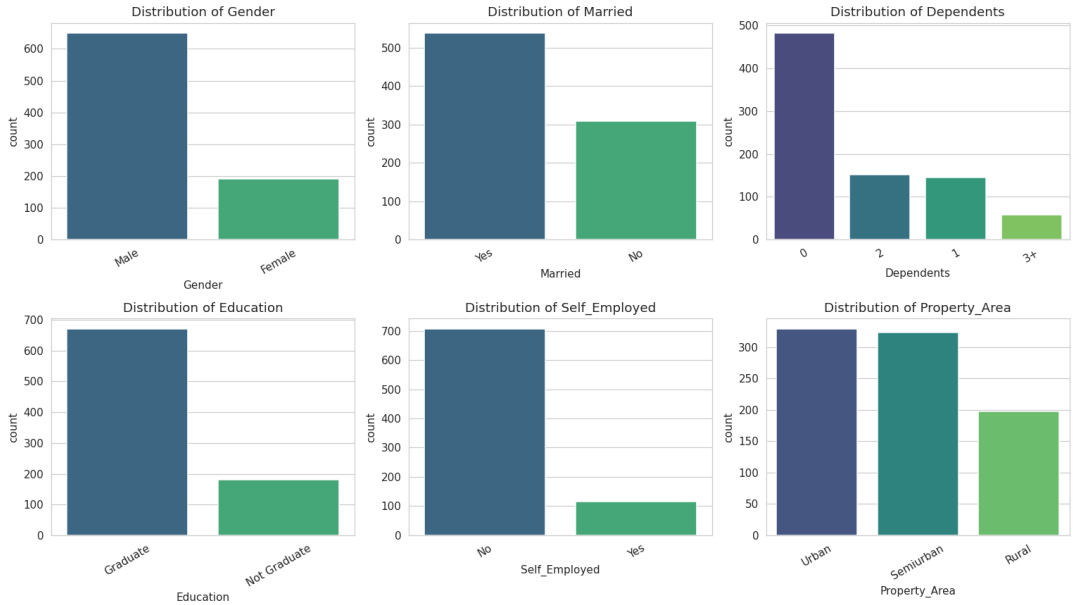


Figure 6: Class distribution of categorical applicant attributes

Inference: Category imbalance (e.g., far more married, graduate applicants with no dependents) is visible across several categorical features; one-hot encoding with a dropped reference level was used to accommodate this without introducing multicollinearity from the dummy-variable trap.

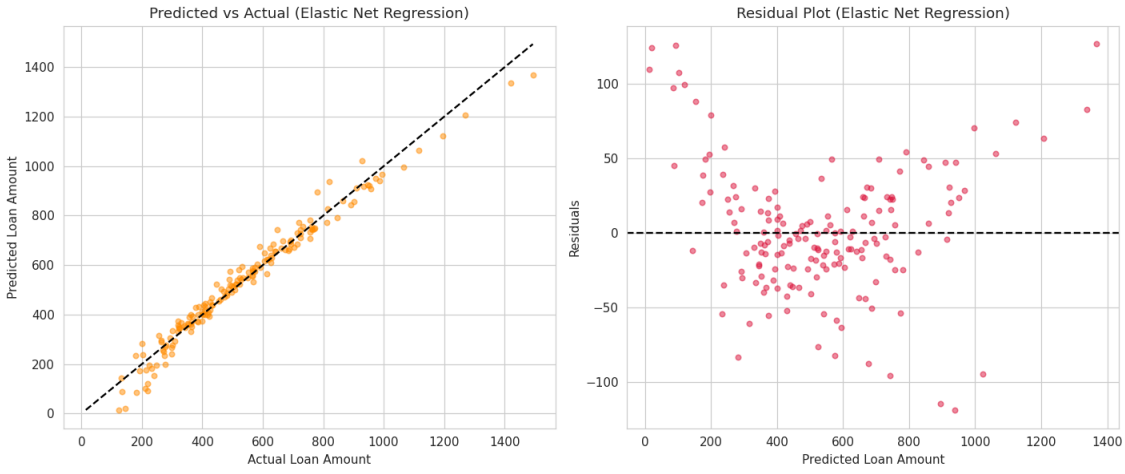


Figure 7: Predicted vs. Actual values and Residual plot for the best model (Elastic Net Regression)

Inference: Predicted values track the actual loan amounts closely along the diagonal reference line, confirming a strong fit (test $R^2 \approx 0.971$). Residuals are scattered randomly around zero with no strong funnel shape, suggesting roughly constant error variance, though a few larger residuals appear at higher predicted values, consistent with the right-skewed target distribution.



Figure 8: Training vs. Validation error (learning curve) for the best model

Inference: Training and validation RMSE converge closely as the training set size increases, with only a small, stable gap remaining. This indicates the model is not significantly overfitting and that additional training data would yield diminishing returns given the current feature set.

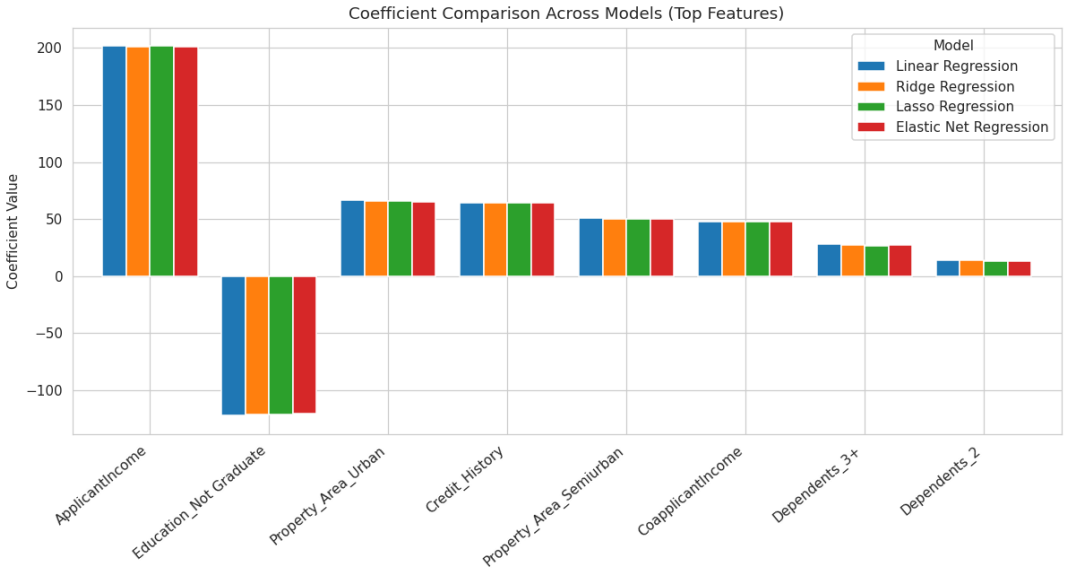


Figure 9: Coefficient comparison across Linear, Ridge, Lasso, and Elastic Net (top features)

Inference: `ApplicantIncome` has by far the largest positive coefficient across all four models (≈ 201), confirming it as the dominant driver of loan amount. Ridge shrinks coefficients only slightly relative to plain Linear Regression, while Lasso and Elastic Net shrink smaller coefficients (e.g. `Dependents_2`, `Dependents_3+`) further toward zero, illustrating Lasso's feature-sparsity behavior.

Table 6: Coefficient Comparison (Top Features)

Feature	Linear	Ridge	Lasso	Elastic Net
ApplicantIncome	201.60	201.30	201.50	201.18
Education_Not Graduate	-122.16	-121.13	-121.66	-120.72
Property_Area_Urban	66.61	65.69	65.56	65.29
Credit_History	64.15	64.08	64.10	64.05
Property_Area_Semiurban	51.22	50.40	50.17	50.02
CoapplicantIncome	47.73	47.59	47.72	47.54
Dependents_3+	28.08	27.55	26.26	27.23
Dependents_2	13.92	13.67	12.89	13.51



Figure 10: Test set performance comparison (MAE, RMSE, R2) across all four models

Inference: All four models perform almost identically on the held-out test set ($R^2 \approx 0.9705$–0.9706), with Elastic Net marginally ahead on RMSE and R2. This suggests the dataset's linear signal is strong and clean enough that regularization mainly stabilizes coefficients rather than materially changing predictive accuracy.

11 Discussion

Across all four models, test-set R^2 is consistently around 0.970–0.971, and the gap between 5-fold cross-validated training performance and test performance is small (e.g. CV RMSE $\approx$ 38.7 vs. test RMSE $\approx$ 43.0), indicating the models generalize well rather than memorizing the training data. Grid search consistently selected small regularization strengths (Ridge $\alpha = 1$, Lasso $\alpha = 0.1$, Elastic Net $\alpha = 0.01$), which is expected given the low multicollinearity observed in the correlation matrix – there is limited redundant information for regularization to compress. The main strength of this experiment is the clear, consistent dominance of `ApplicantIncome` as a predictor across every model, which makes the results easy to interpret. A limitation is that, because the numeric predictors are only weakly correlated with each other, the practical benefit of Ridge/Lasso/Elastic

Net over plain Linear Regression is small on this particular dataset; the advantage of regularization would likely be more pronounced on a noisier dataset with more correlated or higher-dimensional features. Potential improvements include engineering an income-to-loan-term ratio feature, exploring polynomial or interaction terms, and testing tree-based models (e.g. Random Forest, Gradient Boosting) as a non-linear comparison baseline.

12 Conclusion

Linear Regression, Ridge, Lasso, and Elastic Net regression models were implemented and evaluated for predicting sanctioned loan amount from applicant demographic and financial features. After 5-fold cross-validated grid search, all four models converged to very similar performance (test $R^2 \approx 0.970$–0.971), with Elastic Net Regression ($\alpha = 0.01$, $l1_ratio = 0.8$) achieving the best test RMSE (42.968) and R2 (0.9706) by a narrow margin over Lasso and Ridge, and all three regularized models slightly outperforming the unregularized baseline. The small, converging training-validation error gap observed in the learning curve confirms the models are neither substantially overfitting nor underfitting. Given the near-identical accuracy, Elastic Net or Ridge would be the preferred choice in practice for their added coefficient stability, at essentially no cost in predictive accuracy, over plain Linear Regression.

13 References

1. Scikit-learn Developers, "Linear Models," *scikit-learn documentation*. https://scikit-learn.org/stable/modules/linear_model.html

2. Scikit-learn Developers, "Tuning the hyper-parameters of an estimator," *scikit-learn documentation*. https://scikit-learn.org/stable/modules/grid_search.html

3. Kaggle, "Predict Loan Amount Data" dataset.